

Supplementary Information

Quadrupole-limited control windows for an atomically addressable ^{49}Ti nuclear-spin qubit

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1 Hamiltonian convention and parameter provenance

The full calculation uses $S = 1/2$, $I = 7/2$ and

$$\frac{H}{h} = \frac{\mu_B}{h} [g_z(B + B_{\text{tip}} \cos \phi)S_z + g_{\perp}B_{\text{tip}} \sin \phi S_x] + \frac{\mu_N}{h} g_N B I_z \quad (1)$$

$$+ A_z S_z I_z + A_{\perp} (S_x I_x + S_y I_y) + \frac{\kappa}{2I(2I-1)} \left[I_z^2 - \frac{1}{2}(I_x^2 + I_y^2) \right]. \quad (2)$$

The fitted STM tip field is applied to the electron Zeeman term and not the nuclear Zeeman term, following the physical convention of the single-atom NMR model [2]. The ^{47}Ti tip parameters are contextual sensitivity inputs rather than measurements on the ^{49}Ti readout junction.

Table 1: Numerical inputs and their role.

Quantity	Value or range	Role
$I(^{49}\text{Ti})$	7/2	isotope property
A_z baseline	130 MHz	^{49}Ti hyperfine scale [1]
A_z comparison	132.1 MHz	^{47}Ti fit [2]
$A_{\perp}/A_z$	0, 0.05, 0.10, 0.20	sensitivity grid
κ_{47}	-56.7 ± 0.8 MHz	^{47}Ti fit [2]
κ_{49}	-46.3738 ± 2.6554 MHz	same-EFG isotope scaling
$ g_N(^{49}\text{Ti}) $	0.314686	numerical nuclear factor
g_z	0.56 ± 0.02	contextual sensitivity range
B_{tip}	67.9 ± 2.1 mT	contextual sensitivity range
ϕ	$5 \pm 2^\circ$	contextual sensitivity range
$g_{\perp}/g_z$	0.5, 1.0, 1.5	sensitivity grid

1.1 Isotope-scaling arithmetic

Stone [4] gives the recommended ground-state moments

$$Q_{47} = 0.302(10) \text{ b}, \quad Q_{49} = 0.247(11) \text{ b}. \quad (3)$$

For the same EFG,

$$R_Q = \frac{Q_{49}}{Q_{47}} = 0.8178808, \quad (4)$$

with relative standard uncertainty

$$\frac{\sigma_R}{R_Q} = \sqrt{\left(\frac{0.011}{0.247}\right)^2 + \left(\frac{0.010}{0.302}\right)^2} = 0.05550. \quad (5)$$

Thus $\kappa_{49} = R_Q \kappa_{47} = -46.37384$ MHz. Combining the κ_{47} fit uncertainty and the quadrupole-moment uncertainty in quadrature gives

$$\sigma_{\kappa_{49}} = \sqrt{(R_Q 0.8)^2 + (56.7 R_Q 0.05550)^2} = 2.6554 \text{ MHz}. \quad (6)$$

The dominant uncertainty in this isotope scaling comes from the nuclear quadrupole moments, not the fitted κ_{47} alone.

2 Explicit derivation of the rigid-comb result

Using $I_x^2 + I_y^2 = I(I+1) - I_z^2$, the diagonal axial quadrupole energy is

$$\frac{E_Q(m_I)}{h} = \frac{\kappa}{4I(2I-1)} [3m_I^2 - I(I+1)]. \quad (7)$$

The signed adjacent transition $m_I \rightarrow m_I - 1$ in electron manifold m_s is therefore

$$\nu_{m_s, m_I} = \gamma_N B + A_z m_s + \frac{3\kappa}{4I(2I-1)} (2m_I - 1). \quad (8)$$

The physical spectroscopy measures $|\nu|$. In the strong-hyperfine regime $|A_z|/2 > |\gamma_N B| + 3S$, all seven $m_s = +1/2$ transitions retain the sign of $A_z/2$ and all seven $m_s = -1/2$ transitions have the opposite signed orientation. Taking the absolute value therefore reverses the order of the latter comb. With $n, m = -3, \dots, 3$,

$$f_n^{(+)} = \frac{|A_z|}{2} + \gamma_N B + nS, \quad (9)$$

$$f_m^{(-)} = \frac{|A_z|}{2} - \gamma_N B - mS, \quad (10)$$

$$S = \frac{3|\kappa|}{2I(2I-1)}. \quad (11)$$

Hence

$$f_n^{(+)} - f_m^{(-)} = 2\gamma_N B + (n+m)S \quad (12)$$

and

$$\Delta_{\text{inter}}^{\min}(B) = \min_{\ell} |\ell S - 2\gamma_N B|. \quad (13)$$

This derivation makes explicit why the common $|A_z|/2$ offset cancels from the observable inter-manifold line separation.

The nearest point of one rigid comb to the other can never lie farther than half a comb spacing, so $\Delta_{\text{inter}}^{\min} \leq S/2$. Since the nearest intra-manifold spacing is S , the inter-manifold constraint is always the tighter one for dual-manifold control.

3 General spin dependence

For the axial convention used above,

$$S_I = \frac{3|\kappa|}{2I(2I-1)}, \quad \Delta_{\text{max}} = \frac{3|\kappa|}{4I(2I-1)}, \quad (14)$$

and

$$B_r^{\text{max}} = \frac{3(2r+1)|\kappa|}{8I(2I-1)|\gamma_N|}. \quad (15)$$

The following dimensionless coefficients illustrate the spin dependence without importing host-specific quadrupole couplings. The central point is that the field prescription is not specific to Ti. Once I , γ_N and the host-dependent κ are known, the same comb construction gives the crossing and maximum fields.

Table 2: Axial comb coefficients for representative high-spin nuclei. The formulas require the host-specific κ and γ_N for numerical fields.

Representative nucleus	I	$S_I/ \kappa $	$\Delta_{\max}/ \kappa $
^{173}Yb	5/2	3/20	3/40
$^{49}\text{Ti}, ^{123}\text{Sb}$	7/2	1/14	1/28
^{209}Bi	9/2	1/24	1/48

4 Full-model sensitivity envelope and validation

The sensitivity grid contains $3 \times 3 \times 3 \times 3 \times 2 \times 4 \times 3 = 1944$ scenarios. It is deterministic and should not be interpreted as a statistical confidence set. The reconstructed implementation was validated against the previous archived values before the low-field extension was accepted:

Field	reconstructed worst (MHz)	archived worst (MHz)	difference (MHz)
1.35 T	0.0541847	0.0541960	-1.13×10^{-5}
1.714 T	1.0927570	1.0927410	$+1.60 \times 10^{-5}$

The low-field search used 1 mT spacing from 0.95 to 1.11 T, followed by 0.2 mT refinement around the coarse maximum. The result is

$$B_{\text{minimax}} = 1.0292 \text{ T}, \quad \min_{\text{grid}} \Delta_{\text{inter}}^{\text{min}} = 1.187772 \text{ MHz}, \quad (16)$$

with a contiguous coarse-grid interval 0.992 to 1.069 T for which the worst sampled separation is at least 1 MHz.

4.1 One-at-a-time sensitivity decomposition

Each documented parameter range was varied independently around the source-backed realization at 1.0292 T. Figure 1 and Table 3 show the resulting span of $\Delta_{\text{inter}}^{\text{min}}$. The full minimax corner combines several extremes,

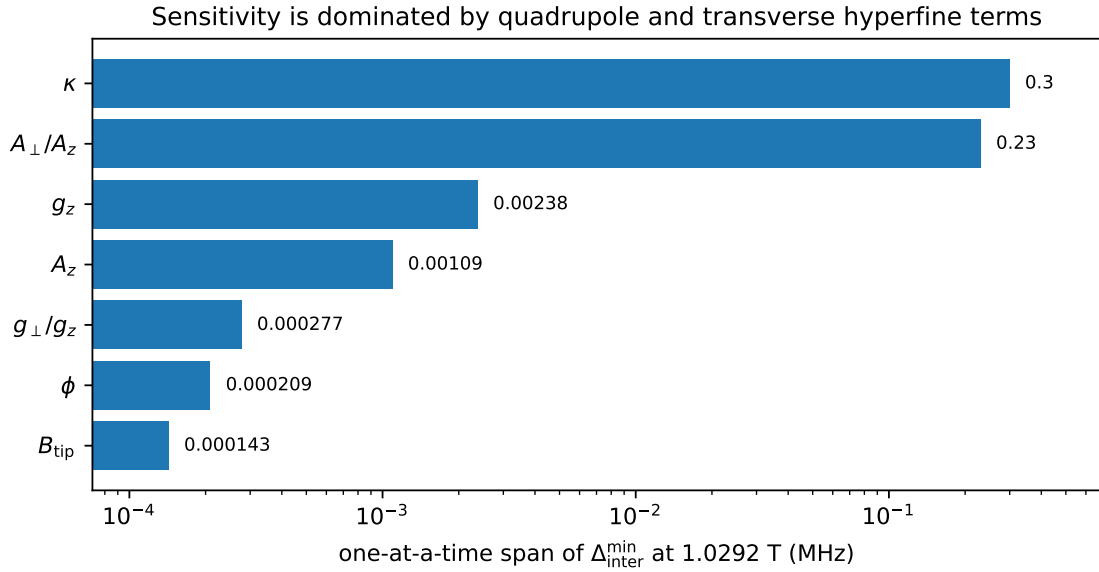


Figure 1: One-at-a-time span of the full-model source-backed separation at 1.0292 T. The quadrupole coupling and transverse hyperfine ratio dominate the sensitivity by more than two orders of magnitude over the tip-field and angular axes.

but the one-at-a-time decomposition shows that the model leverage is concentrated in κ and $A_{\perp}/A_z$.

Table 3: One-at-a-time spans at 1.0292 T.

Parameter axis	span in $\Delta_{\text{inter}}^{\text{min}}$ (MHz)
κ	0.29969
$A_{\perp}/A_z$	0.23041
g_z	0.00238
A_z	0.00109
$g_{\perp}/g_z$	0.000277
ϕ	0.000209
B_{tip}	0.000143

5 Fourteen-tone control construction and numerical integration

For a static electron manifold, the projected nuclear basis has seven adjacent links. The physical waveform contains the seven frequencies from each of the two electron manifolds. A common physical amplitude scale f_R is applied to all tones. For transition m , the interaction-picture coupling is proportional to

$$f_R d_m, \quad d_m = \langle m-1 | I_x | m \rangle, \quad (17)$$

so the Clebsch-Gordan amplitudes needed for an I_y rotation arise automatically from the projected I_x matrix elements. Each tone phase is $\arg(d_m) + \pi/2$. This is the multitone SU(2) construction used in high-spin nuclear control [5, 6, 7].

The target is $U_{\text{target}} = \exp[-i(\pi/2)I_y]$ and $t_g = 1/(4f_R)$. The interaction-picture 8×8 propagator is integrated independently in both electron manifolds under the same waveform using DOP853 with relative tolerance 2×10^{-8} and absolute tolerance 2×10^{-10} . The maximum time step is restricted by the gate time and by the largest detuning below 15 MHz. For $d = 8$, the average gate fidelity is

$$F_{\text{avg}} = \frac{\text{Tr}(U^{\dagger}U) + |\text{Tr}(U_{\text{target}}^{\dagger}U)|^2}{d(d+1)} \quad (18)$$

[8]. The reported control metric is the minimum of the two electron-manifold fidelities.

The source-backed realization is $A_z = 130$ MHz, $\kappa = -46.3738411$ MHz, $g_z = 0.56$, $B_{\text{tip}} = 67.9$ mT, $\phi = 5^\circ$, $A_{\perp}/A_z = 10/130$ and $g_{\perp}/g_z = 1$. Dense scans use 0.25 kHz spacing around the first 99% crossing.

Table 4: Representative source-backed worst-manifold fidelities.

f_R (kHz)	1.35 T	1.0292 T	1.714 T
5	0.988043	0.999647	0.999475
10	0.941427	0.997627	0.997439
20	0.545803	0.992079	0.990802
35	0.198257	0.967298	0.976625

The interpolated thresholds are 4.8150, 21.9289 and 21.6133 kHz at 1.35, 1.0292 and 1.714 T, respectively. The non-monotonicity close to the threshold is visible in Fig. 2 of the main text. These coherent revivals motivate the first-crossing definition rather than the largest isolated sampled passing point.

A previous diagnostic normalized the two optimized curves by the matched source-backed $\Delta_{\text{inter}}^{\text{min}}$. Because those two separations differ by less than 1%, this comparison does not provide a sufficient lever arm to establish a scaling law. It is retained in the archived exploratory analysis but is not used as a publication-facing claim. A genuine f_R^{99} versus $\Delta_{\text{inter}}^{\text{min}}$ scaling test requires additional control propagations at intermediate fields.

6 Electron occupation and switching specification

The single-shot experiment was performed at $T = 0.4$ K and reports significant thermal population of the excited electron state [1]. Using only the external-field source-backed Zeeman splitting at 1.0292 T,

$$\frac{g_z \mu_B B}{h} = 8.07 \text{ GHz}, \quad \frac{h\nu}{k_B} = 0.387 \text{ K}, \quad (19)$$

which gives the two-level Boltzmann estimate

$$p_{\text{exc}} = \frac{1}{1 + e^{0.387/0.4}} \simeq 0.275. \quad (20)$$

Including the source-backed longitudinal tip field gives about 0.263. These numbers are orientation estimates rather than a fit to the non-equilibrium STM electron populations.

The same experiment cites an electron lifetime of order 100 ns on the atom, while $f_R^{99} = 21.93$ kHz corresponds to $t_g = 11.4 \mu\text{s}$. If the static-manifold propagation is to be interpreted as a gate without switching, an exponential switching time τ_e must satisfy

$$1 - e^{-t_g/\tau_e} \ll 1. \quad (21)$$

Numerically, $P_{\text{switch}} < 0.1$ requires $\tau_e > 108 \mu\text{s}$, and $P_{\text{switch}} < 0.01$ requires $\tau_e > 1.14$ ms. These are control specifications, not demonstrated electron lifetimes. The pulsed readout protocol demonstrates that DC and RF excitation can be turned off between probe pulses [1]; whether a control configuration can also suppress electron switching to the required scale remains an experimental question.

If the electron is prepared and remains in one manifold, the waveform contains seven rather than fourteen resonant tones and the inter-manifold ceiling $S/2$ no longer binds. The leading-order spacing relevant to adjacent transitions is then S . No seven-tone gate-rate threshold is inferred here because it requires a dedicated propagation.

7 Coherence, linewidth and next measurements

No measured ^{49}Ti nuclear T_2 is assumed. Under the adjacent-level pure-dephasing model of the archived analysis, $T_{2,\text{adj}}/t_g$ is approximately 10^3 , 10^4 and 10^5 for 99%, 99.9% and 99.99% target fidelities. At 20 kHz, these correspond to 12.6 ms, 126 ms and 1.26 s.

A future selectivity figure of merit is $Q_{\text{sel}} = \Delta_{\text{inter}}^{\text{min}}/\Gamma_{\text{NMR}}$, where Γ_{NMR} is a linewidth measured under the relevant drive and junction conditions. The fit precision reported in the ^{47}Ti single-atom NMR preprint is not used as a linewidth [2]. The highest-leverage measurements are therefore: (i) direct ^{49}Ti NMR through 0.99 to 1.07 T; (ii) direct ^{49}Ti quadrupole and transverse-hyperfine extraction on the target junction; (iii) time-domain nuclear Rabi calibration; (iv) Ramsey and echo T_2 ; (v) electron switching rates during the control interval; and (vi) adsorption-site or tip-configuration dependence of the EFG.

8 Reproducibility package

The version 2.0.0 repository-ready folder accompanying this revision contains the low-field 16-level scan, validation records, dense control sweeps, one-at-a-time sensitivity data, field-quadrupole design map, scripts, publication figures and SHA-256 checksums. The earlier Zenodo release remains available as version 1.0.0 at DOI [10.5281/zenodo.21969250](https://doi.org/10.5281/zenodo.21969250); the version 2.0.0 release should be created under the same concept record before final publication.

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